INCH-POUND
MIL-PRF-32023
6 May, 1998

PERFORMANCE SPECIFICATION
ADAPTER KIT - GRAVITY FEED, FOR 5-GALLON MILITARY PLASTIC FUEL CAN

1. SCOPE

1.1 Scope. This document covers the performance and acceptance requirements for a Gravity Feed Adapter (GFA) which mates with the standard military 5- gallon plastic fuel can. The GFA and can assembly are inverted to provide a gravity fed fuel supply for military tent heaters (see 6.1).

2. APPLICABLE DOCUMENTS

Not applicable to this specification.

3. REQUIREMENTS

3.1 First article test. When specified, see 6.2, the GFA shall undergo a first article test in accordance with 4.1.1.

3.2 Conformance. Items furnished under this specification shall be subject to the conformance inspection in accordance with 4.1.2.

3.3 Operating requirements.

3.3.1 Fuel flow rate. The GFA shall allow a minimum fuel flow rate from an inverted five gallon fuel can of 200 cubic centimeters (cc)/minute of JP8 fuel in all operating temperatures. No more than 325 cc's of fuel shall remain in the can after draining.

3.3.2 Sealing. There shall be no leakage of fuel from the GFA, when connected to a full five gallon military plastic fuel can, upon inversion for operation, during operation, or if accidentally tipped over during transport. The maximum torque required for a positive seal shall not exceed 120 in·lbs in all operating temperatures.

Beneficial comments, recommendations, additions, deletions, and any pertinent data which may be used in improving this document should be addressed to : U.S. Army Natick Research Development and Engineering Center, ATTN: SSCNC-WEC, Natick, MA 01760-5018

AMSC N/A FSC 7240

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3.3.3 Attachment. The GFA shall be quickly and easily attached and removed from the can by hand while wearing cold weather gloves. No tools shall be required for attachment or removal. The GFA shall mate with the male threads on the military five gallon plastic fuel can without damaging fuel can or GFA threads. Attachment for operation shall require no more than the following three steps: 1. Attach adapter to fuel can, 2. Attach fuel line to adapter, 3. Invert fuel can on fuel can stand.

3.4 Support or ownership requirements.

3.4.1 Storage. The GFA body size shall not exceed 5.5 inches in diameter and 3.0 inches in height for storage inside a space heater.

3.4.2 Marking. Lettering shall be permanent and shall not wear off due to contact with fuel or normal wear associated with use of the GFA. The information on the GFA shall contain the following at a minimum:

a. Manufacturer's name and address or cage code.
b. Manufacturer's part number or NSN.

3.5 Environmental requirements.

3.5.1 Water tightness. The GFA shall prevent water penetration into the fuel can during operation, transport, or storage under all environmental conditions.

3.5.2 Extreme temperatures. The GFA and materials shall remain operable in temperatures from 65° F down to -60° F.

3.5.3 Fuel compatibility. All GFA materials (body, hose, coupler, O-rings, gaskets, etc..) shall be compatible with liquid fuels (JP8, Kerosene, Diesel).

3.5.4 Corrosion resistance. The GFA, fittings, and materials shall not rust.

3.6 Interface and interoperability requirements.

3.6.1 Fuel supply hose and connector. The GFA shall be equipped with a 6-inch long, 3/8-inch inside diameter flexible hose with a automatic shut-off female coupler. The coupler shall mate with male fitting Parker Hannifin Corp. Part no. BH8CP or equivalent which shall be supplied with the adapter when specified (see 6.2).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Conformance inspection (see 4.1.2)

4.1.1 First article inspection. When a first article inspection is required (see 3.1 and 6.2), the GFA shall undergo the tests and examinations outlined in Table I. Sampling for the first article inspection shall be as specified in the contract or purchase order.

4.1.2 Conformance inspection. Conformance inspections shall include tests 4.3.3, 4.4.2, 4.5.2, 4.5.3, and 4.6.1. Sampling for conformance inspections shall be as specified in the contract or purchase order.

TABLE I Requirement and Verification Outline

Requirement	Requirement Paragraph	Verification Paragraph	Requirement	Requirement Paragraph	Verification Paragraph
Operating requirements	**3.3**	**4.3**	**Environmental requirements**	**3.5**	**4.5**
Flow rate	3.3.1	4.3.1	Water tightness	3.5.1	4.5.1
Sealing	3.3.2	4.3.2	Extreme temperature	3.5.2	4.5.2
Attachment	3.3.3	4.3.3	Fuel compatibility	3.5.3	4.5.3
Support or ownership requirements	**3.4**	**4.4**	Corrosion resistance	3.5.4	4.5.4
Storage	3.4.1	4.4.1	**Interface and interoperability requirements**	**3.6**	**4.6**
Marking	3.4.2	4.4.2	Fuel supply hose and connector	3.6.1	4.6.1

4.2 Verification methods. Verification methods can include visual examination, measurement, testing, simulation, modeling, engineering evaluation, component properties analysis, certification and similarity to previously approved or previously qualified designs.

4.2.1 Verification alternatives. The contractor may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance (see 6.4).

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4.3 Operating requirements.

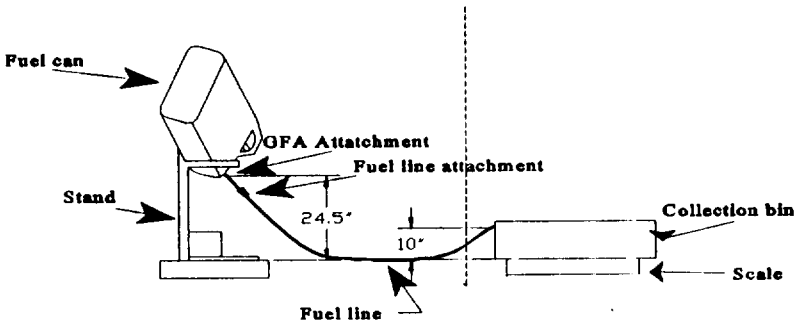


Figure 1. Flow test set up

4.3.1 Flow rate test. Cold soak a GFA and full five gallon can (NSN 7240-01-337-5269) with JP-8 fuel (NSN 9130-01-031-5816) in an ambient temperature of -40°F (+/- 2°F) for a period no less than 4 hours. While maintaining a temperature of -40°F (+/- 2°F), put the GFA on the can and invert on a stand. Connect a 3/8 inch I.D. fuel line to the GFA which shall empty into a collection bin. The outlet of the fuel line and height of the can shall be as shown in figure 1. The collection bin shall be placed on a digital scale (minimum resolution 0.05 kg) which shall be kept in surrounding temperatures within manufacturer's recommendation for scale operation. Record the increase in the collection bin weight every 2 minutes until flow has stopped and record final weight of the fuel can. Compare the final weight of can to the weight of an empty can and verify no more than 325 cc of fuel remain in the can. Verify the average flow rate from the can does not drop below 200 cc/min.

4.3.2 Sealing inspection. During test of 4.3.1, measure the torque required to provide a positive seal between the fuel can and GFA. At the end of the test measure the torque required to remove the GFA. Torque shall be measured from the center of the GFA using an appropriate fixture and a calibrated torque wrench which measures in inch pounds. The torque wrench shall be kept at room temperature until immediately before measurements are taken. At no time shall the torque exceed 120 in·lbs. Observe GFA/can assembly during flow test of 4.3.1. At no time shall there be any fuel leaks from the GFA/fuel can assembly. With a GFA attached to a full 5-gallon fuel can, tip assembly on it's side and verify there is no fuel leakage.

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4.3.3 Attachment inspection. Verify that the GFA mates with the external threads on the 5 gallon fuel can (NSN 7240-01-337-5269). While wearing cold weather gloves, verify the GFA can be easily attached to, and removed from the can without the use of tools. Verify the GFA is fully sealed by inverting a full fuel can after attachment and observing for leaks. Attach and remove the GFA a minimum of ten times. There shall be no cross threading encountered or difficulties attaching the GFA. There shall be no thread wear or deformation on either the fuel can or the GFA threads. Verify the GFA can be attached and removed in 10 seconds or less and that there are no more than 3 steps required for GFA attachment and setup in operating position.

4.4 Support or ownership requirements.

4.4.1 Storage inspection. Verify through measurement that the GFA body dimensions are less than 5.5 inches diameter and less than 3 inches in height.

4.4.2 Marking inspection. Verify the adapter has the minimum information required in 3.4.2.

4.5 Environmental requirements.

4.5.1 Water tightness test. Weigh an empty can with the GFA attached and record (minimum scale resolution shall be .01 kg). Subject one can assembly in the upright position and one in an inverted position to 4 in/hr of rain directed at a 45 degree angle for 32 minutes (8 minutes per each side or the can). Re-weigh the GFA/can assembly and verify there was no water penetration.

4.5.2 Extreme temperature test. During test of 4.3.1, manually bend the GFA fuel supply hose in a 90 degree angle, then back 180 degrees in the opposite direction, then straight in less than 15 seconds. Repeat this test 5 times. There shall be no damage to the hose. Verify that the fuel line fitting, Parker P/N BH8CP, is easily attachable to the female coupler on the GFA. If the GFA requires disassembly for storage, verify disassembly and assembly can be accomplished under the extreme temperatures without tools while wearing cold weather gloves.

4.5.3 Fuel compatability inspection. Inspect material certification of GFA, hose, gaskets, and fittings for compatability with diesel and JP-8 fuels.

4.5.4 Corrosion resistance inspection. Inspect material certification of GFA, hose, and fittings for non-rusting properties.

4.6 Interface and interoperability requirements.

4.6.1 Fuel supply hose and connector inspection. Verify presence of a 6-inch by 3/8-inch inside diameter fuel supply hose with a female automatic shut off coupler. Verify the mating fitting, Parker PN BH8CP, or equivalent, mates with the GFA and is included when specified (see

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6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Departments System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The GFA is intended for use with the 5 gallon plastic fuel can NSN 7240-01-337-5269 to provide a gravity feed fuel supply capability in temperatures ranging from 65° F down to -60° F. When attached to the can, the can and GFA assembly are inverted and placed on the fuel can stand (NSN 7240-01-318-5222). The GFA is intended for use with the following heaters:

Heater, Space, Radiant Type, Portable, 45,000 Btuh, (H45) - NSN 4520-01-329-3451
Heater, Space, Radiant, Type II, Liquid Fuel (M41) - NSN 4520-00-927-4214
Heater, Space, M1950 (Yukon) - NSN 4520-00-287-3353
Heater, Space, Radiant, Liquid and Solid Fuel, Type I, SHA - NSN 4520-01-444-2364
Heater, Space, Radiant, Liquid and Solid Fuel, Type II, SHS - NSN 4520-01-444-2375
Heater, Space, Convective - NSN 4520-01-431-8927

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required (see 3.1)
c. Sampling plan, lot size and sample size (see 4.1.1 and 4.1.2)
d. Packaging requirements (see 5.1)
e. When a male fitting is supplied with the adapter (see 3.6.1 and 4.6.1)

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. The documents should

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also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Verification alternatives. Contracting documents should provide guidance to offerors regarding the submission of alternatives to specified verification methods (See 4.2.1)

6.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the materials meet or exceed the operational and maintenance requirements and promote economically advantageous life cycle costs.

6.6 Subject term (key word) listing.

Tent Heating
Diesel Supply Line

Custodians:

Army - GL
Navy - YD1
Air Force - 99

Preparing activity:

Army - GL

Review activities:

Army - MD1
Navy - MC
Air Force - 84
GSA - FSS

Project No: 7240-0275

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	MIL-PRF-32023	980506

3. DOCUMENT TITLE
Adapter Kit, Gravity Feed, For 5-Gallon Military Plastic Fuel Can

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial (508) 233-5175	(2) AUTOVON 256-5175
c. ADDRESS *(Include Zip Code)* ATTN: SSCNC-WEF Kansas Street Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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